TEXORA · LECTURE SHEET

Process Control in Carding

Carding control centres on nep removal, sliver evenness and waste discipline at the closest settings in the mill. This sheet covers nep counting through the line, the setting regime, waste norms, and the production–quality trade-off that governs doffer speed decisions.

YM-4101 · PROCESS CONTROL IN SPINNING · TOPIC 4 · SPECIALIZATION — YARN MANUFACTURING

1. Neps — the card's report card

A nep is a small entangled knot of fibres, often with trash or immature fibre at its core. Neps are counted per gram of material at each stage (AFIS or web inspection). The pattern through the line is characteristic: blowroom action *creates* some neps while it cleans, and the card must remove far more than it receives. A card that merely holds the nep level steady is failing.

WORKED EXAMPLE — NEP BALANCE THROUGH OPENING AND CARDING

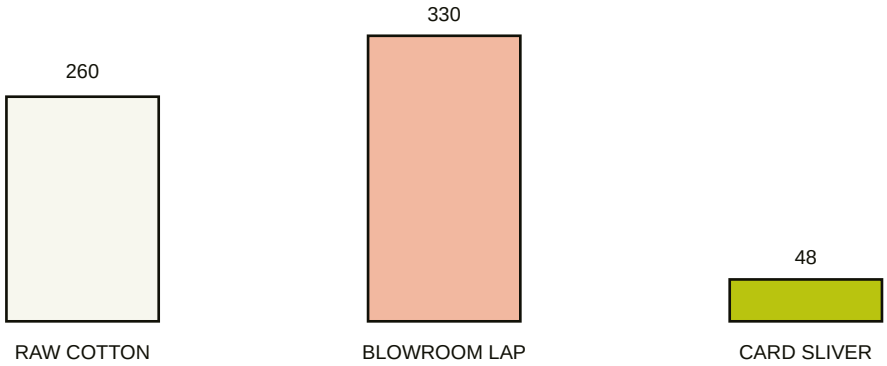
Nep counts (per gram): raw cotton in the mixing 260; blowroom lap 330; card sliver 48.

The blowroom increased neps by (330 − 260)/260 = 26.9% — the cost of beating, to be minimised by gentle settings (Topic 3) but never zero.

Card nep removal, measured against its own feed = (330 − 48)/330 × 100 = **85.5%**.

Overall change, raw cotton to sliver = (260 − 48)/260 × 100 = **81.5%** reduction.

Control reading: if sliver neps rise while lap neps are unchanged, the fault is in the card (settings, clothing, overload). If lap neps rise too, the cause is upstream — the mixing (immature fibre) or blowroom settings.



Neps per gram (bar heights to scale). Carding removes 85.5% of the neps it receives.

Figure 1 — Nep balance of the worked example: created in the blowroom, removed at the card.

2. Settings control

Carding is a game of thousandths of an inch. Settings open up in service through wear, vibration and cleaning, and are verified with feeler gauges on a schedule — and after every clothing change or grinding.

SETTING	TYPICAL VALUE (COTTON)	IF TOO WIDE	IF TOO CLOSE
Cylinder – flats	0.20–0.25 mm	Nep removal collapses, sliver U% rises	Clothing damage, fibre rupture, fire risk
Taker-in – feed plate	0.15–0.25 mm	Tufts enter unopened, nep load rises	Fibre damage at entry, mote knife choking
Cylinder – doffer	0.10–0.15 mm	Poor transfer, web patches, cylinder loading	Doffer clothing damage
Taker-in – cylinder	0.15–0.20 mm	Fibre carried back, licker-in waste rises	Clothing strike risk

3. Waste discipline at the card

WASTE STREAM	NORM (SHARE OF FEED)	DIAGNOSTIC MEANING WHEN HIGH
Licker-in droppings	2–3%	Feed too trashy, or mote-knife/grid setting too open
Flat strips	2–3%	Genuinely dirty material — or flats stripping good fibre when settings/clothing are poor

Total card waste	4–6%	Compare with feed trash: waste should follow trash, not exceed it by a widening margin

Flat strips deserve a fibre check, not just a weighing: strips rich in long, spinnable fibre mean the flats are harvesting the product. Strip rate is also compared card to card — one card stripping far above its neighbours on the same lap is a machine fault, not a material fact.

4. Sliver quality targets

- **Sliver U%** — typically held at or below about 3.0–3.5 for medium counts on modern cards; a rising trend precedes visible faults.
- **Sliver weight CV** — follows lap weight CV (Topic 3) plus the card's own contribution; checked by the cut-and-weigh test each shift.
- **Trash in sliver** — residual particles per gram; the card is the last heavy cleaning point before combing or drawing.
- **Production rate discipline** — the rate calculated in YM-3207 Topic 2 (13.0 kg/h at 85% for 0.15 Ne sliver) is the quality-approved rate. Raising doffer speed raises output per hour but cuts cleaning time per kilogram; nep and U% results must be re-proved at any new rate before it becomes standard.

5. Card-to-card comparison

The most sensitive routine tool is the comparison of identical measures across all cards on the same feed: sliver weight, U%, neps, strip rate. The department average describes the material; a single card standing off the line describes a machine. Control charts (Topic 1) are kept per card for this reason.

Glossary

Nep — a small knot of entangled fibres, often around a trash or immature-fibre core. **AFIS** — the instrument that counts neps, trash and fibre length properties on small samples. **Clothing** — the wire-covered surfaces of cylinder, doffer and flats. **Flat strips** — waste stripped from the revolving flats. **Mote knife** — the knife at the taker-in that deflects trash into the droppings. **Loading** — fibre embedding into clothing until carding action is lost. **Grinding** — the periodic sharpening of card clothing.

Key points

• Neps rise in the blowroom (+26.9% in the example) and must fall at the card (−85.5% of those received).
• Diagnose by location: sliver neps up with lap neps steady = card fault; both up = material or blowroom.
• Cylinder–flat setting (0.20–0.25 mm) is the single most important control on the machine.
• Waste norms: licker-in 2–3%, flat strips 2–3%, total 4–6% of feed — and strips must be checked for good fibre.
• Production increases must be re-proved against nep and U% results before becoming standard.

Practice

1. Lap neps are 300/g and card sliver neps are 75/g. Calculate the card's nep removal efficiency. If the target is 85%, what is the maximum permissible sliver nep count on this lap?
2. Two cards on identical lap give sliver nep counts of 44/g and 71/g. State the likely class of cause and the first three checks on the worse card.
3. A card's flat strips rise from 2.4% to 3.8% of feed while feed trash is unchanged. Give two possible causes and how you would distinguish them.

Answer key

1. Removal = (300 − 75)/300 × 100 = **75%**. At 85% removal the sliver may carry 300 × 0.15 = **45 neps/g** at most — the card is underperforming and needs investigation.
2. A machine-specific (assignable) cause, since the material is common. First checks: cylinder–flat setting (gauged), clothing condition/loading and grinding state, and taker-in setting with mote-knife condition.
3. (1) Flats stripping good fibre — settings too close or clothing damaged; distinguish by analysing the strips for long-fibre content. (2) Material reaching this card is dirtier than the department average (e.g. lap from one blowroom line); distinguish by testing trash in the laps actually fed to this card against other laps.